

Rohit Kumar

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SUMMARY

B.Tech ECE student at IIIT Jabalpur specializing in full-stack web development with React, Node.js, TypeScript, and PostgreSQL; shipped production systems with JWT/RBAC authentication, payment webhook integration, and LLM API integration. Competitive programmer with 1,200+ problems solved and LeetCode contest rating 1520.

EDUCATION

PDPM IIIT Design and Manufacturing

Jabalpur, MP

B.Tech in Electronics and Communication Engineering – CGPA: 7.7/10

2024 – 2028

- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Digital Systems
- Qualified **JEE Advanced** — among top 1.5% of 1.4 million JEE Main candidates nationwide

PROJECTS

Creator Dashboard | *React 18, Node.js, Express.js, PostgreSQL, JWT, Razorpay, Tailwind CSS* [Live](#) | [GitHub](#)

- Developed and deployed a full-stack creator monetization web application using React 18, Node.js, Express.js, and PostgreSQL; integrated Razorpay payment gateway with real-time **webhook processing** and **HMAC-SHA256 signature verification** to automate content access control; deployed split-stack on Vercel, Render, and Neon PostgreSQL
- Architected **JWT authentication** with **Role-Based Access Control (RBAC)** across 5 RESTful API endpoints with automatic token refresh on role elevation; designed a payment lifecycle system (pending → active → cancelled) using **PostgreSQL partial unique index** constraints and JOIN-based authorization middleware to prevent duplicate subscriptions and privilege escalation

LeetCode Analyzer | *TypeScript, React, Express.js, PostgreSQL, Drizzle ORM, Groq LLM, Judge0* [Live](#) | [GitHub](#)

- Developed and deployed a full-stack LeetCode submission analysis application using React, TypeScript, Node.js, and Express.js in a **monorepo architecture**; delivers automated correctness verification, time and space complexity analysis, algorithmic pattern recognition, and a two-stage Socratic hint system without user authentication
- Engineered a **custom multi-language code execution harness generator** parsing LeetCode GraphQL API signatures into runnable Python, C++, and Java programs executed in a self-hosted **Judge0 sandbox**; integrated **Groq LLM API** with structured JSON contract and PostgreSQL query caching via Drizzle ORM pre-seeded with ~3,000 problems; enforced per-IP rate limiting to control inference costs

CF Enhancer | *JavaScript, Vite, Chart.js, Chrome Extension Manifest V3, Codeforces REST API* [GitHub](#)

- Developed a **Google Chrome Extension (Manifest V3)** replacing the default Codeforces profile page with a competitive programming analytics dashboard featuring contest rating history, problem difficulty distribution, topic-level weakness detection, and a GitHub-style yearly activity heatmap using **Chart.js** visualizations
- Integrated **Codeforces Public REST API** through a dedicated **background service worker** for MV3 compliance and separation of concerns; implemented **Single Page Application (SPA) navigation detection** via **MutationObserver Web API** to automatically re-initialize the dashboard on every client-side route transition

COMPETITIVE PROGRAMMING

LeetCode — 571 problems solved | Contest Rating **1520** | Ranked **1,717 / 37,970** (Top 4.5%) in Weekly Contest 508

Codeforces — 238 problems solved | Rating **1186**

CodeChef — 438 problems solved | Rating **1401** (2★)

Aggregate: 1,200+ Data Structures and Algorithms problems solved across platforms with strong coverage in Dynamic Programming, Graphs, Trees, and Binary Search

TECHNICAL SKILLS

Languages: C++, TypeScript, JavaScript, SQL, HTML/CSS

Frameworks & Libraries: React 18, Node.js, Express.js, Drizzle ORM, Tailwind CSS, Chart.js

Databases: PostgreSQL, Drizzle ORM, Neon Serverless PostgreSQL

Developer Tools: Git, GitHub, Linux, Vite, Vercel, Render

Concepts: Data Structures and Algorithms, Object-Oriented Programming, RESTful API Design, JWT Authentication, Role-Based Access Control (RBAC), Webhook Integration, GraphQL API, LLM API Integration, Chrome Extension Development (Manifest V3), Monorepo Architecture, Rate Limiting, CI/CD